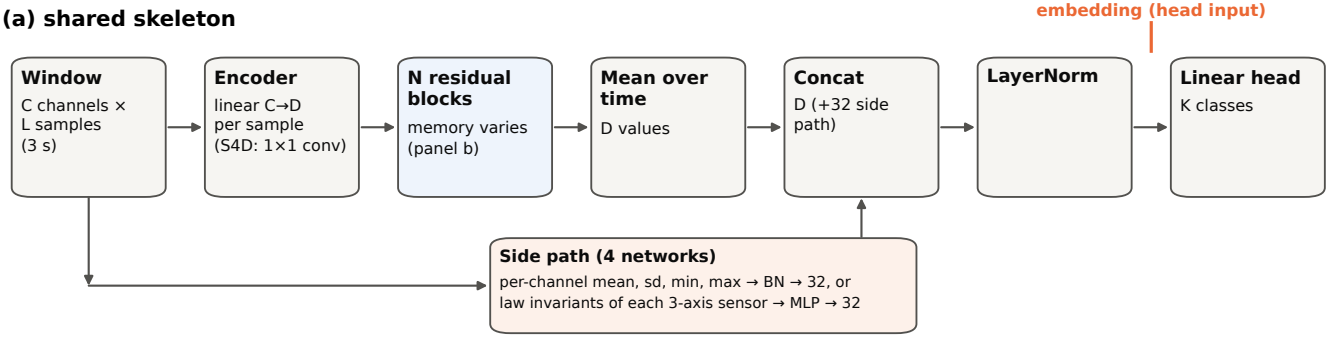
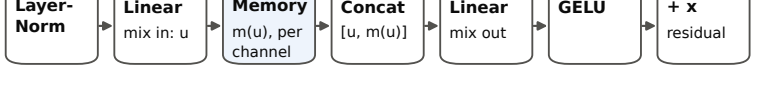
(a) shared skeleton



(b) residual block



recurrence: $r_t(\tau) = \sigma((\varepsilon - \|z_t - z_{t-\tau}\|)/T)$ and $r_t(\tau)\, r_{t-1}(\tau)$, ten lags
play operator: $y_t = \min(\max(y_{t-1}, u_t - r), u_t + r)$, four widths r ($r = 0$: no memory)
quantile tracker: $q_t = q_{t-1} + \eta(\tau - \mathbf{1}[u_t < q_{t-1}])$, four levels τ
S4D block: LayerNorm, diagonal state-space convolution + skip, GELU, 1×1 conv, GLU, + x
Mamba block: LayerNorm, in-projection, depthwise conv, SiLU, selective scan, gate, out-projection, + x

(c) the ten networks (reference input of gate G1)

network	memory + side path	D, N, parameters
RecurrenceNet	recurrence	32, 3, 24,675
S4D	S4D (diagonal SSM)	64, 2, 27,675
HysteresisNet	play operators	32, 3, 23,579
QuantileNet	quantile trackers	32, 3, 20,763
Mamba	Mamba (S6)	32, 2, 21,147
No memory	none (width 0)	32, 3, 19,995
RN + stats	recurrence + channel stats	32, 3, 26,451
RN + laws	recurrence + law invariants	32, 3, 27,553
Laws (no mem.)	none (plain) + law invariants	48, 3, 19,033
Stats (no mem.)	none (plain) + channel stats	48, 3, 17,931